MIL-STD-841B(USAF)

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MILITARY STANDARD

MARKING OF AIRCRAFT AND MISSILE PROPULSION SYSTEM PARTS FABRICATED FROM CRITICAL HIGH TEMPERATURE ALLOYS

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DEPARTMENT OF DEFENSE

WASHINGTON, D.C. 20330

Marking of Aircraft and Missile
Propulsion System Parts Fabricated from
Critical High Temperature Alloys

MIL-STD-841B(USAF)

1. This Military Standard is mandatory for use by all Departments and Agencies of the Department of Defense.

2. Recommended corrections, additions or deletions should be addressed to Headquarters AFLC, Attention: MMOPD, Wright-Patterson Air Force Base, Ohio 45433.

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MARKING OF AIRCRAFT AND MISSILE PROPULSION SYSTEM PARTS FABRICATED FROM CRITICAL HIGH TEMPERATURE ALLOYS

1. SCOPE.

1.1 This document establishes physical item marking requirements for aircraft and missile propulsion system parts and assemblies fabricated from critical high temperature alloys. The requirements of this standard are in addition to the identification marking required by MIL-STD-130.

2. REFERENCED DOCUMENTS.

2.1 The following document in effect on the date of invitation for bid forms a part of this document:

STANDARD

Military

MIL-STD-130 Identification Marking of U.S. Military Property

(Copies of standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. DEFINITIONS.

3.1 Critical Alloys. For the purpose of this standard, Critical Alloys shall be interpreted to mean high temperature alloy specifications containing at least one of the following ratios of critical materials:

20% or more nickel (e.g. AISI Type 310)

25% or more chromium (e.g. AISI Type 446)

10% or more cobalt

80% or more titanium

4. GENERAL REQUIREMENTS.

4.1 Aircraft and Missile Propulsion System parts to be marked.

4.1.1 All compressor and turbine blades or vanes.

4.1.2 All parts and assemblies which contain at least 50% of an alloy described in paragraph 3.1 or a combination of such alloys.

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4.1.3 Parts currently being produced. Items already in stock by contractor, government or supplier need not be marked.

4.2 Marking Methods. The marking methods shall meet all the requirements of MIL-STD-130 except as stated in paragraphs 4.2.1, 4.2.2, and 4.2.3 below.

4.2.1 Marking must be permanent and legible after use. On items which cannot be permanently marked, the material designator will be (a) stamped or stencilled on the part or (b) marked on a tag, attached to each part or assembly or on the package for the item.

4.2.2 The permanent marking of item shall not affect the performance or have deleterious effect on the item. If the item cannot be marked without deleterious effect on the item, the marking shall be (a) stamped or stencilled on the part or (b) marked on a tag or on the package as specified in paragraph 4.2.1.

4.2.3 Inseparable assemblies shall be marked as parts. Component parts of inseparable assemblies need not be marked.

5. DETAIL REQUIREMENTS.

5.1 Parts. Parts shall be marked with the material designation of the critical alloy contained in the part (e.g. AISI 310).

5.2 Inseparable Assemblies. Inseparable assemblies shall be marked with the alloy or combination of critical alloys contained in the assembly (e.g. AISI 310/HS25).

6. INTENDED USE. Permanent identification of the critical high temperature alloys contained in parts and assemblies of propulsion systems for aircraft and missiles provides a positive tool for true and accurate segregation of valuable and strategic materials during disposal processes. This document shall be applied to all contracts or orders covering propulsion systems (engines) for aircraft and missiles.

NOTICE: Copies of this standard for military use may be obtained as indicated in the foreword to the Index of Military Specifications and Standards.

Copies of this standard may be obtained for other than official use by individuals, firms, and contractors from the Superintendent of Documents, U.S. Government Printing Office,

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